Jay Singh

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Professional Summary

Computer Science undergraduate specializing in backend engineering and distributed systems with Java, Spring Boot, and cloud architecture. Developed Strangle (strangle.in), a production real-time voice/video platform powered by WebRTC, Redis, RabbitMQ, and AWS EC2. Experienced in architecting low-latency matchmaking engines, event-driven microservices, real-time signaling protocols, and automated CI/CD pipelines.

Technical Skills

Languages: Java (21), Python, SQL, JavaScript
Backend Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, WebSockets (STOMP), JWT, OAuth 2.0
Distributed Systems & Real-Time: WebRTC, Redis (Sorted Sets, Lua Scripts, Distributed Locks), RabbitMQ, Coturn (STUN/TURN)
Databases & Cloud Storage: PostgreSQL, MySQL, MinIO (S3-Compatible Object Storage)
Cloud, DevOps & Tooling: AWS EC2, Docker, Docker Compose, GitHub Actions CI/CD, Caddy (Reverse Proxy/TLS), Git, Postman, Linux
Core CS & Practices: Data Structures & Algorithms, System Design, OOP, SOLID, API Design, Unit Testing (JUnit, Mockito)

Projects

Strangle – Real-Time Voice/Video Platform | *Spring Boot, WebRTC, Redis, RabbitMQ, Docker, AWS* **Live: strangle.in**

- Architected a high-concurrency matchmaking engine using **Java 21, Spring Boot**, and **Redis (Lua scripts, Sorted Sets)**, achieving atomic queue pairing, verified campus domain isolation, and sub-second successive partner discovery without race conditions.
- Engineered end-to-end **WebRTC** audio/video calling with cross-platform interoperability across **Next.js** (Web) and native **Android**, utilizing **WebSockets (STOMP)** for SDP/trickle ICE signaling, **Coturn (STUN/TURN)** for NAT traversal, and a mutual two-way video consent handshake.
- Built an asynchronous, event-driven messaging layer with **RabbitMQ** and **PostgreSQL**, integrating **MinIO (S3-compatible)** presigned URLs for media transfers and automated **Spring cron workers** for zero-leak ephemeral data shredding and media lifecycle cleanup.
- Integrated a decoupled **Python (FastAPI)** Trust & Safety microservice via RabbitMQ/REST to perform automated toxicity and NSFW media moderation, pairing with real-time user reporting and distributed strike enforcement.
- Containerized and deployed the distributed system on **AWS EC2** using **Docker Compose**, **Caddy** (reverse proxy with automated Let's Encrypt TLS/WebSocket tunneling), and a multi-stage **GitHub Actions CI/CD** pipeline featuring automated smoke tests and selective zero-downtime rebuilds.

Amazon-Inspired E-Commerce Backend | *Spring Boot, MySQL, Redis, Docker* **GitHub**

- Developed an e-commerce backend covering product catalog, inventory, cart, wishlist, and order-management workflows.
- Implemented JWT-based authentication and role-based authorization using Spring Security to protect REST APIs.
- Integrated Stripe Payment Intents and webhooks for payment processing and checkout workflows.
- Used Redis caching and database indexing to improve data access; containerized application dependencies with Docker Compose.

Expense Tracker – Full-Stack Application | *Spring Boot, React, JWT, MySQL* **GitHub**

- Built an expense-management application with JWT authentication, Google OAuth 2.0, and user-specific expense records.
- Designed layered REST APIs using Controller-Service-Repository architecture, Spring Data JPA, MapStruct DTOs, and validation.
- Implemented expense analytics, CSV export, global exception handling, and API documentation using Swagger/OpenAPI.

Education

Chhatrapati Shahu Ji Maharaj University, Kanpur **2023 – 2027**
Bachelor of Technology in Computer Science

- **Relevant Coursework:** Data Structures and Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming, Computer Networks

Certifications & Achievements

TCS iON NQT-IT 2026: Scored 81.22% overall (Cognitive: 83.60%; Java Programming: 82.39%).

LeetCode: Solved 450+ problems covering data structures, algorithms, and problem-solving.